

ForceDelta-VLA: Supplementary Material

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I. TRAINING AND ARCHITECTURE DETAILS

The temporal force-conditioned teacher is fully fine-tuned on eight NVIDIA A100 GPUs. All real-robot inference and runtime measurements use a single 48 GB NVIDIA RTX 4090D GPU. The teacher is trained for 50,000 steps with AdamW at a peak learning rate of 2×10^{-5} . It predicts $H=50$ actions at 30 Hz from the latest 100 ms of the 100 Hz wrench stream, and flow-matching inference uses $N_{\text{flow}}=10$ steps.

After freezing the teacher, we train the rank-32 null-mode adapter for 10,000 steps at a learning rate of 10^{-4} . The visual-language prefix is pooled into $M=16$ intent tokens. The residual policy comprises a causal force encoder and one pre-norm attention block with separate force and staleness heads. It predicts $K=5$ corrections and is trained for 30,000 steps at a learning rate of 3×10^{-4} under replayed asynchronous schedules. The final implementation record will specify the intent-token pooling rule and the embedding used to mask the force token for the staleness head.

The action representation and rotational residual composition are defined in Sec. III-A of the main paper. At deployment, force-conditioned and staleness corrections are bounded independently in the normalized pose-action space before composition with the reference action. The final implementation record will specify the normalization statistics and per-head correction bounds.

A. Null-Mode Optimization

The learned null token and null-specific adapter are active only for null-mode queries. With the force-conditioned teacher parameters θ fixed, the null-mode parameters ϕ are optimized using the teacher's original flow-matching objective,

$$\mathcal{L}_{\text{null}} = \mathbb{E}_{\mathcal{D}, \eta, \epsilon} [\ell_{\text{FM}}(T_{\theta, \phi}^{\text{null}}, A_{t:t+H-1}^*, \eta, \epsilon)], \quad (1)$$

where η is flow time and ϵ is the initial flow noise. Full-force and null-force predictions used to construct a residual target share the same ϵ .

B. Residual-Loss Weighting

For a K -step residual chunk, the normalized temporal weights are

$$w_j = \frac{\beta^{j-1}}{\sum_{m=1}^K \beta^{m-1}}, \quad 0 < \beta \leq 1. \quad (2)$$

Thus, $\beta < 1$ emphasizes near-term corrections, $\beta = 1$ weights all steps equally, and $\sum_j w_j = 1$ keeps the loss scale stable as K changes. The loss is a mean-squared error in normalized pose-action coordinates.

II. ASYNCHRONOUS PACKET EXECUTION

A. Reference Packets

A reference query k , issued at t_k from state S_k , publishes the immutable packet $(A_{T,k,1:H}^{\text{null}}, Z_k^{\text{intent}}, S_k, t_k, \bar{t}_k)$ when inference completes at $\bar{t}_k$. Its samples carry intended execution times

$$t_{k,n}^{\text{ref}} = t_k + (n-1)/f_A, \quad n = 1, \dots, H. \quad (3)$$

Samples whose intended times precede $\bar{t}_k$ are skipped for direct execution; the cached chunk itself is not truncated. For $t \geq \bar{t}_k$, the reference pose $A_k^{\text{ref}}(t)$ is obtained by timestamp-based interpolation over the original timestamped chunk.

B. Residual Packets

A residual inference packet q , initiated at t_q^F , reads the latest reference packet k_q together with $(\mathcal{F}_{t_q^F}, S_{t_q^F})$. It completes at $\bar{t}_q^F$ with force-conditioned and staleness correction chunks. Their samples are timestamped as

$$t_{q,j}^F = t_q^F + (j-1)/f_A, \quad j = 1, \dots, K. \quad (4)$$

Step j is held over $[t_{q,j}^F, t_{q,j}^F + 1/f_A)$; correction samples are not interpolated. At control time t , the executor selects the most recently initiated packet that has completed and remains valid, together with its active step:

$$q^*(t) = \arg \max_{q: \bar{t}_q^F \leq t < \bar{t}_q^F + K/f_A} t_q^F, \quad (5)$$

$$j(t; q^*) = 1 + \lfloor (t - t_{q^*}^F) f_A \rfloor.$$

Any prefix that expires during inference is therefore skipped, while the remaining valid steps execute sequentially until a newer packet completes. Within the selected interval, the runtime quantities in Eq. (1) of the main paper are

$$k = k_{q^*},$$

$$A_k^{\text{ref}}(t) \equiv A_{k_{q^*}}^{\text{ref}}(t_{q^*, j(t; q^*)}^F), \quad (6)$$

$$\Delta \hat{A}^{(\cdot)}(t) = \Delta \hat{A}_{q^*, j(t; q^*)}^{(\cdot)}.$$

The two corrections are bounded independently and composed in normalized delta-action coordinates. The composed command remains referenced to $S_{k_{q^*}}$ and is converted to an absolute robot command using the mapping $\mathcal{R}$ defined in the main paper. A packet that completes after $t_q^F + K/f_A$ is rejected. If no valid residual packet remains, the executor invokes the platform's safe hold or abort policy. The final implementation record will specify gripper sampling and behavior beyond the last valid reference timestamp.